

Autonomous Enterprise Framework (AEF) - Information System Design

Table 1: Design Activities & Key System Questions

Design Activity	Key Question	AEF System Implementation
Design Application Components	What are the key parts and how will they interact?	6-Layer Architecture: UI, Meta-Agent, Orchestration, Knowledge Lake, External Integration, and Governance.
Design the User Interface	How will users interact with the system?	3 Roles: Entrepreneurs (Guided Strategy), Investors (Read-only Analytics), and Compliance (Audit Trails).
Design the Database	How will data be captured and stored?	
		Triple-Storage: Structured (MongoDB), Semantic (Vector DB), and Immutable logs for auditing.
Design Software Classes	What internal structure ensures efficiency?	Modular Microservices with Agent-Internal Roles, Event-Driven Task Queues (Redis), and Containerization.
Describe the Environment	Interaction with external tech?	Deep integration with Web Search, Financial (Stripe/Banking), and Social APIs for market sentiment.

Table 2: Key Information System Parts

Component	Detailed Specifications
User Interface	Web Dashboard for Entrepreneur view; Admin Panel for Internal oversight; Investor view for analytics.
Meta-Agent Layer	Strategic Core: Opportunity detection, Task planning, and Agent spawning decisions.
Knowledge Lake	Shared Memory: Vector DB for strategy; Log store for auditable reasoning; MongoDB for profiles.
External Integration	Web APIs for research; Financial APIs for payments; Social APIs for trend analysis.
Governance Layer	Budget control system, Security controls, and Compliance monitoring (2026 Standards).

Table 3: Interaction & Deployment Flow

Step	From -> To	Action Performed
1	User -> UI -> Meta-Agent	Goal submission and strategy generation.
2	Meta-Agent -> Orchestrator	Task breakdown and assignment to sub-agents.
3	Orchestrator -> Sub-Agents	Digital labor execution with budget caps.
4	Sub-Agents -> Knowledge Lake	Writing and reading reasoning logs and results.
5	Orchestrator -> UI Dashboard	Final reports and progress visualisations.